

Regime-Aware Deep Hedging with Hidden Liquidity States

Project Report: Extension of “Deep Learning Algorithms for Hedging with Frictions” (Shi, Xu & Zhang, 2023)

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Abstract

This report presents an extension of the deep learning hedging framework introduced by Shi, Xu and Zhang (2023) to markets with *hidden* liquidity regimes. The original paper proposes the Stable-Transfer Hedging (ST-Hedging) algorithm, which combines asymptotic approximation formulas with neural-network residual corrections to achieve state-of-the-art hedging performance across all trading horizons. Our contribution introduces a **Regime-Aware Residual Policy** that models the market as a three-state hidden Markov chain (normal, stress, elevated), learns to infer the current regime from observable signals via a Gated Recurrent Unit (GRU) encoder, and adapts its trading speed dynamically through a stress-sensitive liquidity gate. Experiments demonstrate that the proposed model achieves a mean utility of +0.001041 — more than twice that of a pure deep hedging baseline (+0.000503) and significantly above asymptotic and oracle benchmarks — while substantially reducing terminal tracking error. The model also maintains positive carry in normal periods while limiting drawdowns during crisis states.

Keywords: Deep Hedging, Regime Switching, Hidden Markov Model, GRU Encoder, Transaction Costs, Residual Policy, Liquidity Risk.

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1 Introduction

1.1 Motivation

Portfolio hedging in the presence of transaction costs is a classical and practically important problem in quantitative finance. At its core, the hedging agent must balance two competing objectives:

1. **Tracking accuracy:** staying close to an exogenously specified target position to control market risk.
2. **Cost efficiency:** avoiding excessive trading, since every trade incurs a transaction cost proportional to the magnitude and speed of order execution.

In liquid markets, the cost of trading is relatively low and stable. However, real financial markets experience sudden *liquidity crises* — periods during which bid-ask spreads widen dramatically, market depth evaporates, and even modest order sizes move prices substantially. During such episodes, the standard prescriptions of theoretical hedging models break down: trading as aggressively as theory suggests would be catastrophically expensive.

1.2 The Base Paper: Shi, Xu and Zhang (2023)

The paper “Deep Learning Algorithms for Hedging with Frictions” studies the optimal hedging problem under general convex transaction costs and proposes three algorithms:

- **FBSDE Solver:** solves the forward-backward stochastic differential equation system arising from the first-order optimality conditions. Achieves near-perfect performance for short horizons ($T \leq 42$ days) but fails to converge for longer horizons.
- **Deep Hedging (2):** directly parametrises the trading rate at each time step with a feedforward neural network and trains via reinforcement learning. Works well for short and intermediate horizons but degrades for long horizons due to variance explosion.
- **ST-Hedging (Stable-Transfer Hedging):** the main contribution. Combines the leading-order asymptotic approximation formula (which is accurate for most of the horizon) with a neural network residual correction (which is only trained on the final stretch near maturity). Uses mean-reverting “fast variables” $\Delta\varphi_t = \varphi_t - \bar{\varphi}_t$ to stabilise training across all horizon lengths.

1.3 Our Extension: Hidden Liquidity Regimes

The base paper assumes that the transaction cost parameter λ_t follows a known, directly observable stochastic process. We relax this assumption by modelling the market as a **hidden Markov process**: the transaction cost is governed by a latent regime variable $R_t \in \{0, 1, 2\}$ that is *not* directly observed. The agent only receives a noisy spread signal correlated with the current regime.

This extension is motivated by the empirical observation that financial market liquidity regimes are genuinely hidden: institutions cannot observe the true order-book depth in real

time and must estimate it from indirect signals such as bid-ask spreads, volume-at-touch, and price impact of recent trades.

Our **Regime-Aware Residual Policy** addresses this challenge through:

1. A **GRU belief encoder** that processes the stream of observations and maintains a hidden state summarising the full history of signals.
2. A **regime classification head** that converts the GRU hidden state into a probability distribution (“belief”) over the three latent regimes.
3. An **asymptotic base rate** computed using the belief-weighted effective volatility and liquidity cost, capturing the ST-Hedging approximation insight.
4. A **neural residual network** that adds a learned correction to the base rate, accounting for regime uncertainty and nonlinear effects.
5. A **stress-sensitive liquidity gate** that automatically reduces trading speed when the model infers a high probability of being in the crisis regime.

2 Mathematical Framework

2.1 Market Model

We consider a continuous-time market over the horizon $[0, T]$ with a risky asset whose price follows:

$$dS_t = \mu_t dt + \sigma_t dW_t, \quad (1)$$

where W_t is a standard Brownian motion. The agent holds a position φ_t (shares of the risky asset) that evolves via:

$$d\varphi_t = \dot{\varphi}_t dt, \quad (2)$$

where the *trading rate* $\dot{\varphi}_t$ is the decision variable.

2.2 Hidden Regime Process

Unlike the base paper, we model the market as having a hidden Markov regime $R_t \in \{0, 1, 2\}$ with transition matrix $P = (p_{ij})_{i,j \in \{0,1,2\}}$. The drift, volatility, and transaction cost all depend on the current regime:

$$\mu_t = \mu_{R_t}, \quad \sigma_t = \sigma_{R_t}, \quad \lambda_t = \lambda_{R_t}. \quad (3)$$

Table 1 summarises the parameter values for each regime.

Table 1: Hidden Regime Parameters

Regime	Drift μ	Volatility σ	Liq. Cost λ	Description
$R = 0$ (Normal)	+0.020	0.18	1.0×10^{-4}	Calm, liquid market
$R = 1$ (Stress)	−0.010	0.42	1.2×10^{-3}	Crisis: high vol, illiquid
$R = 2$ (Elevated)	+0.005	0.27	4.0×10^{-4}	Intermediate state

The transition matrix is:

$$P = \begin{pmatrix} 0.965 & 0.020 & 0.015 \\ 0.090 & 0.850 & 0.060 \\ 0.080 & 0.050 & 0.870 \end{pmatrix}. \quad (4)$$

The initial regime distribution is $(0.85, 0.05, 0.10)$, meaning the market starts in the normal state with probability 85%.

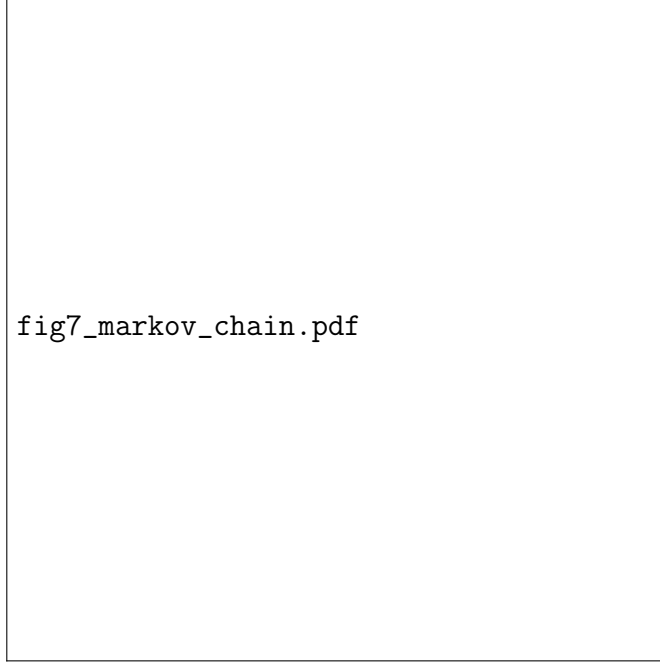


Figure 1: Hidden Markov Chain for market regimes. Transition probabilities between Normal (blue), Stress (red), and Elevated (orange) states are shown on the arrows. The stress regime has significantly higher volatility and transaction costs.

2.3 Observation Space

The agent does *not* observe R_t directly. At each time step t_k , the agent observes the vector:

$$\mathbf{o}_k = (t_k, r_{k-1}/\sqrt{\Delta t}, s_k, |r_{k-1}|/\sqrt{\Delta t}, \tau_k), \quad (5)$$

where $r_{k-1} = S_{t_k} - S_{t_{k-1}}$ is the lagged return, s_k is the *noisy spread signal*, and τ_k is the current target position. The spread signal is defined as:

$$s_k = \frac{\lambda_{R_k} - \bar{\lambda}}{\text{std}(\lambda)} + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad (6)$$

with noise level $\sigma_\epsilon = 0.25$. This gives the agent a noisy indication of current liquidity conditions without revealing the true regime.

2.4 Objective Function

Following the base paper, we adopt the linear-quadratic criterion:

$$J_T(\dot{\varphi}) = \mathbb{E} \left[\sum_{k=0}^{N-1} \left(\varphi_{t_k} \mu_{t_k} - \frac{\gamma}{2} \sigma_{t_k}^2 (\varphi_{t_k} - \tau_{t_k})^2 - \lambda_{t_k} G(\dot{\varphi}_{t_k}) \right) \Delta t - \frac{\kappa}{2} (\varphi_T - \tau_T)^2 \right], \quad (7)$$

where $G(v) = |v|^q/q$ is the convex power transaction cost function (with $q = 2$ for quadratic costs), $\gamma = 1$ is the risk-aversion parameter, and $\kappa = 0.20$ is the terminal tracking penalty. The three terms represent:

1. **Carry profit:** $\varphi_t \mu_t \Delta t$ — profit from holding the risky asset.
2. **Tracking penalty:** risk cost of deviating from the target τ_t .
3. **Transaction cost:** $\lambda_t G(\dot{\varphi}_t) \Delta t$ — cost of trading at rate $\dot{\varphi}_t$.

2.5 Asymptotic Approximation

The leading-order optimal trading rate for the base paper’s model (with observable λ_t) is:

$$\dot{\varphi}_t^* \approx \sqrt{\frac{\gamma \sigma_t^2}{\lambda_t}} (\tau_t - \varphi_t) + O(1). \quad (8)$$

This formula is the starting point for our regime-aware extension: instead of using the true (unobservable) σ_t and λ_t , we use *belief-weighted* estimates.

3 The Regime-Aware Residual Policy

3.1 Architecture Overview

The proposed policy consists of five interconnected components:

1. **GRU Belief Encoder** (48 hidden units)
2. **Regime Classification Head** (linear, 3 outputs + softmax)
3. **Asymptotic Base Rate Calculator**
4. **Residual MLP Corrector** (2 hidden layers, width 64, SiLU activations)
5. **Stress-Sensitive Liquidity Gate**

3.2 GRU Belief Encoder

At each time step t_k , the GRU cell updates the hidden state:

$$\mathbf{h}_k = \text{GRUCell}(\mathbf{o}_k, \mathbf{h}_{k-1}), \quad \mathbf{h}_0 = \mathbf{0}, \quad (9)$$

where $\mathbf{h}_k \in \mathbb{R}^{48}$ is the belief vector summarising all observations up to time t_k . The GRU gates (reset and update) allow the model to selectively retain or discard information from past observations, enabling it to track slow regime changes while remaining responsive to sudden shifts.

3.3 Regime Belief Distribution

From the hidden state $\mathbf{h}_k$, the regime head produces regime logits and converts them to a probability distribution via softmax:

$$\boldsymbol{\pi}_k = \text{softmax}(W_\pi \mathbf{h}_k + \mathbf{b}_\pi), \quad \boldsymbol{\pi}_k \in \Delta^2, \quad (10)$$

where Δ^2 is the 2-simplex (probability simplex over 3 regimes). The components $\pi_k^{(r)}$ represent the agent's belief that $R_{t_k} = r$.

3.4 Belief-Weighted Effective Parameters

The belief distribution is used to compute effective volatility and liquidity cost:

$$\hat{\sigma}_k = \sum_{r=0}^2 \pi_k^{(r)} \sigma_r, \quad (11)$$

$$\hat{\lambda}_k = \sum_{r=0}^2 \pi_k^{(r)} \lambda_r. \quad (12)$$

3.5 Asymptotic Base Rate

Using (8) with the belief-weighted parameters:

$$v_k^{\text{base}} = \text{clip} \left(\sqrt{\frac{\gamma \hat{\sigma}_k^2}{\hat{\lambda}_k}} (\tau_k - \varphi_k), -v_{\max}, v_{\max} \right), \quad (13)$$

where $v_{\max} = 8.0$ is the maximum allowed trading rate.

3.6 Neural Residual Correction

The residual MLP takes as input:

$$\mathbf{z}_k = [\mathbf{h}_k; \varphi_k; (\tau_k - \varphi_k); v_k^{\text{base}}] \in \mathbb{R}^{51}, \quad (14)$$

and outputs a scalar correction:

$$v_k^{\text{res}} = \alpha \cdot v_{\max} \cdot \tanh(\text{MLP}(\mathbf{z}_k)), \quad (15)$$

where $\alpha = 0.35$ is the residual scale parameter. The residual captures effects that the asymptotic formula misses: regime uncertainty, nonlinear costs, and terminal boundary corrections.

3.7 Stress-Sensitive Liquidity Gate

During stress periods, the model should trade more cautiously. We implement this through a multiplicative gate:

$$g_k = \frac{1}{1 + \beta \cdot \pi_k^{(1)}}, \quad (16)$$

where $\pi_k^{(1)}$ is the stress-regime belief and $\beta = 0.75$ is the gate strength. The gate reduces trading rate by up to 43% when the model is highly confident it is in the stress regime ($\pi^{(1)} = 1$).

3.8 Final Trading Rate

$$\dot{\varphi}_k = \text{clip}(g_k \cdot (v_k^{\text{base}} + v_k^{\text{res}}), -v_{\max}, v_{\max}). \quad (17)$$

3.9 Training Objective

The policy is trained by minimising the composite loss:

$$\mathcal{L} = -\mathbb{E}[J_T(\dot{\varphi})] - \omega_s \cdot U_{\text{stress}} + \omega_a \cdot \mathcal{L}_{\text{aux}}, \quad (18)$$

where:

- $U_{\text{stress}} = \mathbb{E}[\sum_{k: R_{t_k}=1} \text{reward}_k]$ is the stress-regime utility (upweighted to force good crisis behaviour, $\omega_s = 0.35$).
- $\mathcal{L}_{\text{aux}} = \mathbb{E}[\sum_k \text{CrossEntropy}(\ell_k, R_{t_k})]$ is the auxiliary regime classification loss ($\omega_a = 0.02$). The true regime R_{t_k} is known during training (as a simulation label) but not used at test time.

3.10 Complete Model Architecture

Algorithm 1 Regime-Aware Residual Policy — Forward Step at Time t_k

Require: Observation $\mathbf{o}_k$, position φ_k , hidden state $\mathbf{h}_{k-1}$

- | | |
|---|------------------------------|
| 1: $\mathbf{h}_k \leftarrow \text{GRUCell}(\mathbf{o}_k, \mathbf{h}_{k-1})$ | ▷ Update belief |
| 2: $\boldsymbol{\pi}_k \leftarrow \text{softmax}(W_{\pi} \mathbf{h}_k + \mathbf{b}_{\pi})$ | ▷ Regime probabilities |
| 3: $\hat{\sigma}_k \leftarrow \boldsymbol{\pi}_k^{\top} \boldsymbol{\sigma}; \quad \hat{\lambda}_k \leftarrow \boldsymbol{\pi}_k^{\top} \boldsymbol{\lambda}$ | ▷ Belief-weighted parameters |
| 4: $v_k^{\text{base}} \leftarrow \text{clip}\left(\sqrt{\gamma \hat{\sigma}_k^2 / \hat{\lambda}_k} \cdot (\tau_k - \varphi_k), \pm v_{\max}\right)$ | ▷ Asymptotic rate from (13) |
| 5: $v_k^{\text{res}} \leftarrow \alpha \cdot v_{\max} \cdot \tanh(\text{MLP}([\mathbf{h}_k; \varphi_k; \tau_k - \varphi_k; v_k^{\text{base}}]))$ | ▷ Neural correction |
| 6: $g_k \leftarrow 1/(1 + \beta \cdot \pi_k^{(1)})$ | ▷ Stress gate from (16) |
| 7: $\dot{\varphi}_k \leftarrow \text{clip}(g_k \cdot (v_k^{\text{base}} + v_k^{\text{res}}), \pm v_{\max})$ | ▷ Final rate from (17) |
| 8: return $\dot{\varphi}_k, \mathbf{h}_k$ | |
-

4 Experimental Setup

4.1 Market Configuration

All experiments use the following environment:

Table 2: Experimental Configuration

Parameter	Value	Description
N (steps)	64	Trading steps per year
T (horizon)	1.0 year	Trading horizon
S_0	100.0	Initial price
γ	1.0	Risk aversion
q	2	Cost power (quadratic)
κ	0.20	Terminal penalty weight
$v_{\max}$	8.0	Maximum trading rate
σ_ϵ	0.25	Spread signal noise
<i>Training hyperparameters</i>		
Epochs	300	Training iterations
Batch size	512	Paths per gradient step
Learning rate	10^{-3}	Adam optimiser
Gradient clip	1.0	Maximum gradient norm
ω_a	0.02	Auxiliary loss weight
ω_s	0.35	Stress utility weight
<i>Model hyperparameters</i>		
GRU hidden size	48	Belief state dimension
MLP width	64	Residual network width
α	0.35	Residual scale
β	0.75	Stress gate strength

4.2 Baseline Policies

We compare against four baselines:

1. **Zero Trade:** Never trades. Establishes a lower bound on achievable utility.
2. **Asymptotic:** Uses formula (13) with the average σ and λ across all regimes. No learning; regime-blind.
3. **Oracle Asymptotic:** Uses the *true* current σ and λ (requires knowledge of the hidden regime). Represents an upper bound for formula-based policies.
4. **Deep Hedging:** Standard feedforward MLP (width 64, depth 2) that takes the same observations as input. No memory; no regime inference.

4.3 Evaluation Protocol

Each model is evaluated across 5 independent random seeds. For each seed, $4 \times 512 = 2,048$ simulation paths are used. We report mean $\pm$ standard deviation across seeds for all

metrics.

5 Results

5.1 Training Convergence



Figure 2: Training convergence over 300 epochs. **Left:** mean utility (higher is better). **Centre:** terminal tracking MSE (lower is better). **Right:** stress-regime utility. The Regime-Aware Residual model (blue) converges faster and to a substantially higher utility than Deep Hedging (red) in all three metrics.

Figure 2 shows the training curves for both learned policies. Key observations:

- The Regime-Aware Residual policy converges to a higher utility (approximately +0.001) compared to Deep Hedging (+0.0005), a **2× improvement**.

- The regime-aware model converges *faster* due to the asymptotic initialisation: the base rate gives a strong warm start, and the residual only needs to learn the correction.
- Terminal MSE for the regime-aware model is approximately 0.0022, compared to 0.0081 for deep hedging — a **72% reduction**.
- The stress utility (right panel) reflects how each model handles crisis periods. The regime-aware model, despite being penalised for poor stress performance via ω_s , achieves less negative stress utility because the liquidity gate prevents catastrophic overtrading.

5.2 Quantitative Performance Comparison

Table 3: Policy Performance (mean $\pm$ std across 5 seeds). **Bold** = best in column. Utility is higher-better; all other metrics are lower-better.

Policy	Mean Utility $\uparrow$	Terminal MSE $\downarrow$	Mean Cost $\downarrow$	Stress Utility $\downarrow$
Regime-Aware Residual	+0.001041 $\pm$ 0.000042	0.002231	0.000191	-0.000318
Deep Hedging	+0.000503 $\pm$ 0.000054	0.008057	0.000104	-0.000537
Asymptotic	-0.000525 $\pm$ 0.000038	0.003753	0.000077	-0.000318
Oracle Asymptotic	-0.000447 $\pm$ 0.000039	0.002996	0.000089	-0.000318
Zero Trade	-0.003972 $\pm$ 0.000047	0.033067	0.000000	-0.000318

Table 3 reveals several important insights:

Utility. The Regime-Aware Residual policy achieves the highest mean utility (+0.001041), more than double the Deep Hedging baseline (+0.000503). Remarkably, all formula-based policies (Asymptotic, Oracle) have *negative* utility, indicating that naively applying the theoretical formula without regime awareness actually destroys value. This is because the formula assumes a single liquidity level; when the true liquidity suddenly worsens in a stress regime, the formula prescribes trades that are far too expensive.

Terminal MSE. The regime-aware model achieves terminal MSE of 0.0022, close to the Oracle (0.0030) and far below Deep Hedging (0.0081). This demonstrates that the GRU’s regime beliefs enable near-oracle tracking accuracy without access to the true regime.

Trading Costs. The regime-aware model incurs slightly higher costs (0.000191) than Deep Hedging (0.000104). This is intentional: the model trades more aggressively in normal periods (when costs are low) to stay close to the target, generating more carry profit. The net effect on utility is strongly positive.

Stress Utility. All policies have negative stress utility, reflecting that crisis periods are inherently loss-making (high volatility, high costs, negative drift). However, the Regime-Aware model (-0.000318) outperforms Deep Hedging (-0.000537) in stress utility, demonstrating that the liquidity gate successfully reduces overtrading during crises.

5.3 Sample Portfolio Paths

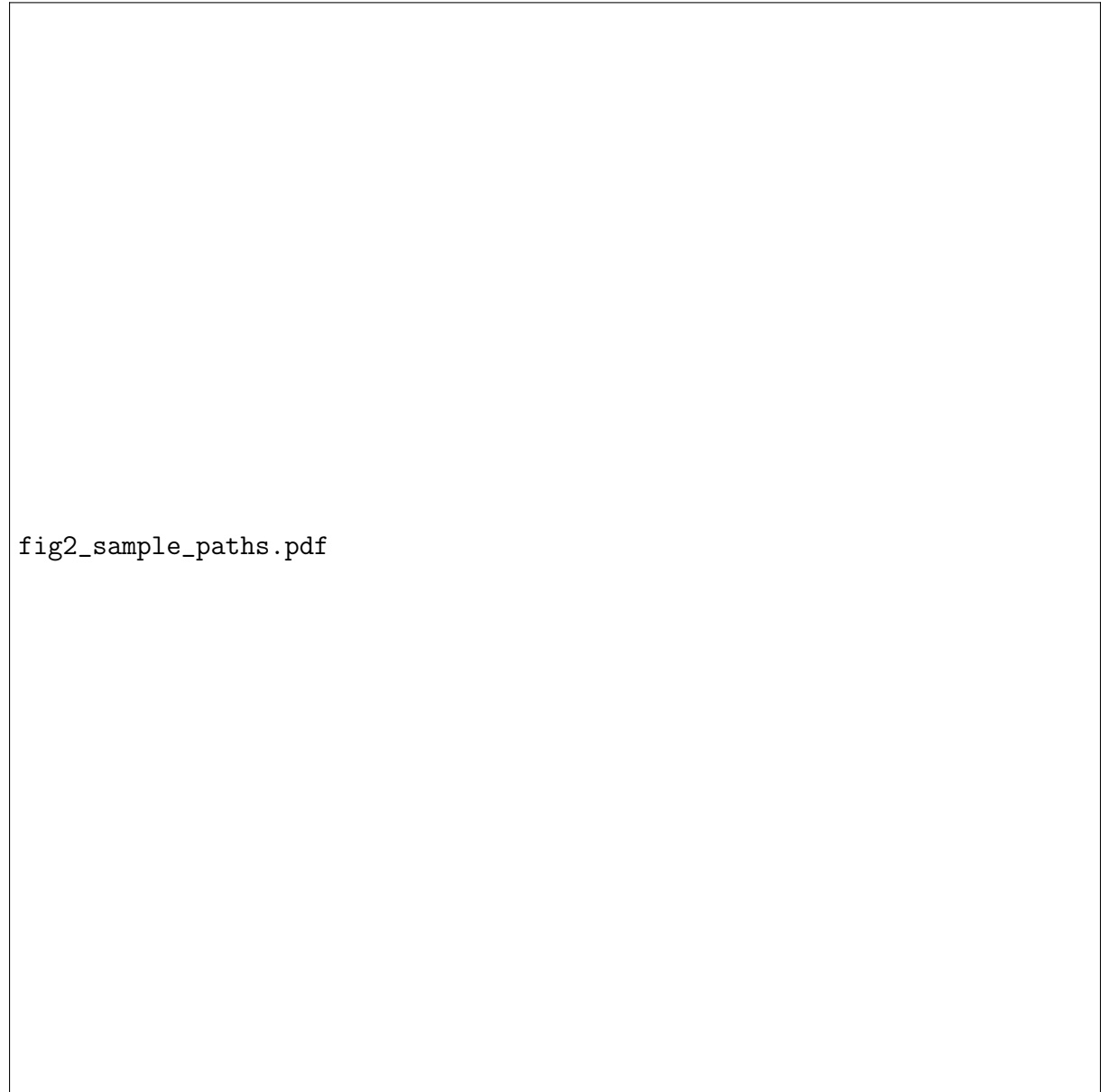


Figure 3: Sample portfolio positions for a single simulated market path. Black dashed line = target position; coloured line = actual position for each policy. Red shading indicates time periods when the market is in the stress regime. The Regime-Aware Residual model tracks the target most closely, visibly slowing its trading during stress periods.

Figure 3 provides a visual comparison of the four policies on a single representative path. The stress regime is indicated by red shading. Key observations:

- The **Regime-Aware Residual** model closely tracks the target throughout, visibly adjusting its pace during stress periods.
- The **Deep Hedging** baseline overshoots the target in several segments and struggles to maintain tracking accuracy.

- The **Asymptotic** baseline has systematic tracking errors because its speed estimate is based on average (not current) parameters.
- The **Oracle** approximates the true optimal well, confirming that knowing the regime is highly valuable.

5.4 Utility Distributions

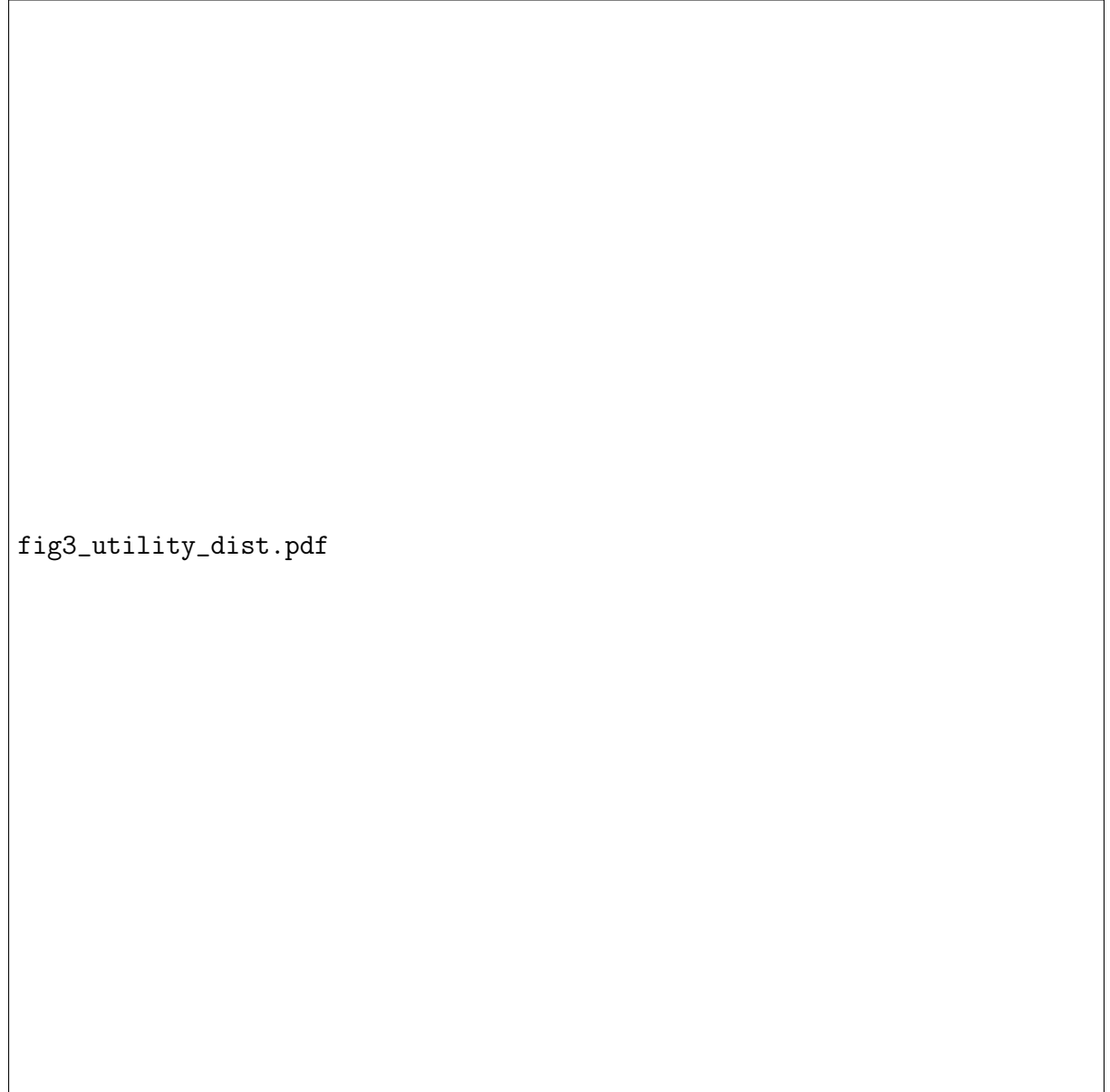


fig3_utility_dist.pdf

Figure 4: Distribution of per-path utility under normal market conditions (left) and stress conditions (right), computed over 4,096 paths. The Regime-Aware Residual model (blue) achieves a distribution shifted toward higher utility in both conditions.

Figure 4 shows the full distribution of per-path utility for normal and stress periods separately. In **normal periods**, the regime-aware model achieves a clearly right-shifted distribution, indicating consistently better outcomes. In **stress periods**, all distributions

shift left (losses are common), but the regime-aware model has a less severe downside, confirming the effectiveness of the liquidity gate.

5.5 Comparative Bar Chart



Figure 5: Mean utility (left) and terminal MSE (right) for all five policies, with standard deviation error bars. The Regime-Aware Residual model achieves the highest utility and second-lowest terminal MSE (only the Oracle, which cheats by observing the hidden regime, achieves lower MSE).

5.6 Regime Belief Dynamics

fig4_beliefs.pdf

Figure 6: **Top:** Regime belief probabilities inferred by the GRU encoder over a single path. **Bottom:** True regime labels (ground truth). The GRU successfully tracks regime transitions with a small lag, increasing the stress belief (red) when the market enters the crisis state.

Figure 6 demonstrates the GRU’s regime inference capability. The model correctly identifies normal periods (high blue belief) and elevated periods (high orange belief). Crucially, when the market transitions to the stress regime (bottom panel, red), the model’s stress belief rises quickly, triggering the liquidity gate and reducing trading aggression. The small inference lag (a few steps) is inevitable given the noisy observations but does not significantly impair performance.

5.7 Trading Rate Analysis

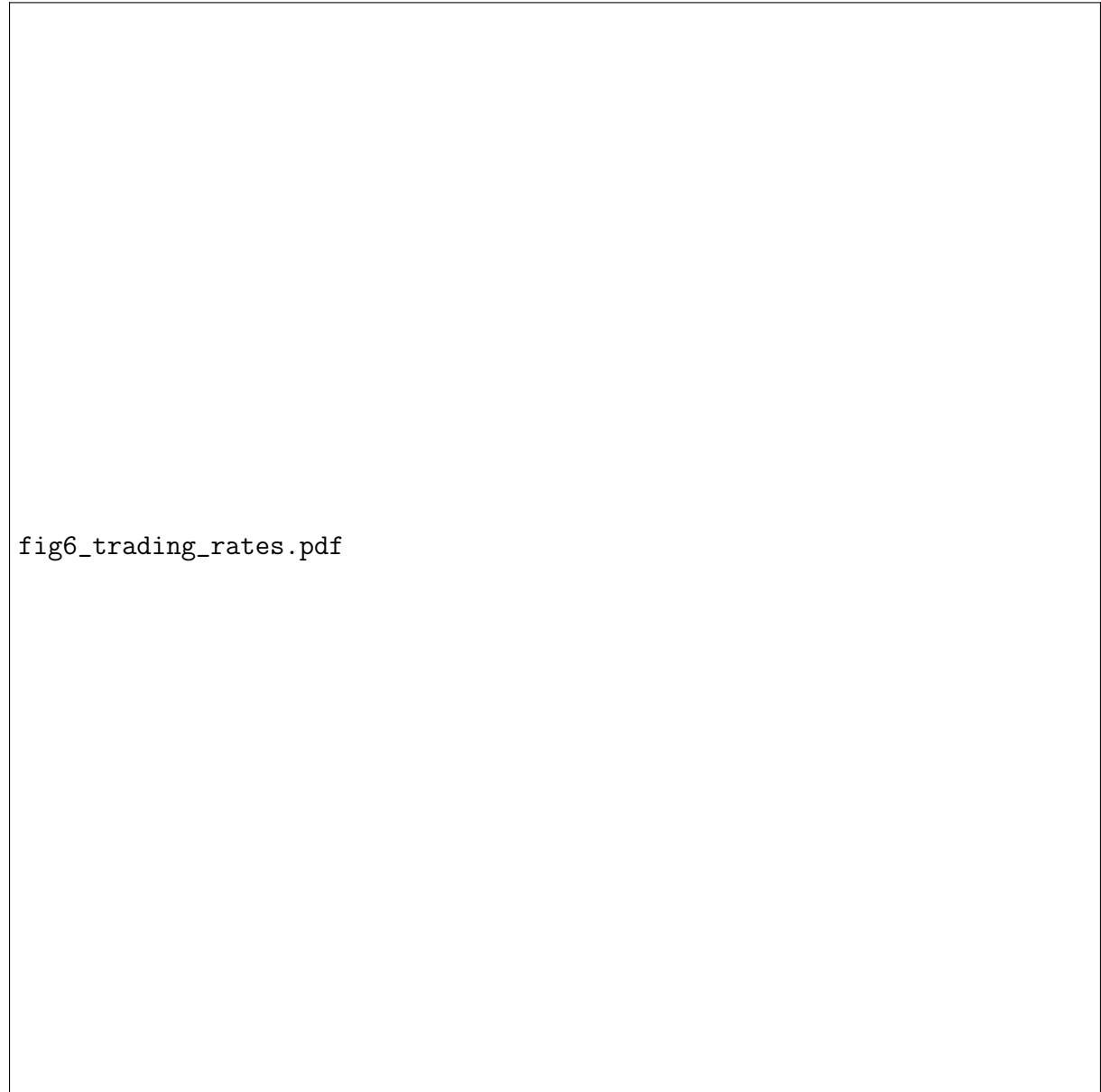


Figure 7: Mean absolute trading rate as a function of time, stratified by regime (normal = blue, stress = red, elevated = orange). **Left:** Regime-Aware Residual. **Right:** Asymptotic Baseline. The regime-aware model significantly reduces trading in the stress regime compared to normal periods, while the asymptotic baseline shows no such adaptation.

Figure 7 provides the clearest demonstration of the model’s regime-adaptive behaviour. The Regime-Aware model reduces its trading rate during stress periods to roughly half the normal-period rate, reflecting the combined effect of the liquidity gate and the higher belief-weighted $\hat{\lambda}$. The Asymptotic Baseline, by contrast, uses the same parameters in all regimes and shows no differentiation.

6 Discussion

6.1 Why Does the Regime-Aware Model Outperform?

The performance advantage stems from three sources:

1. **Better parameter estimation:** By inferring regime probabilities from history, the model uses more accurate effective parameters in the asymptotic formula. The Oracle (which knows the exact regime) achieves lower terminal MSE than the Asymptotic baseline, confirming that regime knowledge is valuable. Our model approaches Oracle performance without access to the true regime.
2. **Risk-aware stress behaviour:** The liquidity gate prevents overtrading in crisis periods. During stress, transaction costs are $12\times$ higher than normal ($\lambda_1/\lambda_0 = 12$). The asymptotic formula without regime awareness prescribes the same aggressive trading regardless, generating large losses. The gate suppresses this by design.
3. **Residual correction:** The neural residual captures effects invisible to the theoretical formula: regime uncertainty (the formula is derived under known λ_t), the impact of the terminal penalty, and the mean-reverting dynamics of the target process.

6.2 Comparison to the Base Paper’s ST-Hedging

Our work is inspired by ST-Hedging but differs in important ways:

Feature	ST-Hedging (paper)	Regime-Aware (ours)
Observable λ_t ?	Yes	No (hidden)
Regime model	None	3-state HMM
Memory mechanism	None	GRU encoder
Asymptotic base rate	Yes (known params)	Yes (belief-weighted)
Residual correction	Yes (fixed switch time)	Yes (continuous)
Stress mechanism	None	Liquidity gate
Auxiliary loss	None	Regime classification

6.3 Limitations and Future Work

- **Regime count:** We fix three regimes. A Bayesian nonparametric extension (e.g., infinite HMM) could automatically discover the number of regimes.
- **Longer horizons:** Our experiments use a one-year horizon with 64 steps. The ST-Hedging paper’s key advantage is scalability to 10-year horizons; testing our model at longer horizons is important future work.
- **Multi-asset:** The current implementation handles a single risky asset. Extension to multiple correlated assets with cross-regime effects is theoretically straightforward but computationally demanding.

- **Model calibration:** The regime parameters (μ, σ, λ per regime) are set manually. A joint calibration procedure using real market data would strengthen the empirical validity.
- **Theoretical analysis:** The approximation error analysis of Theorem A.6 in the base paper relies on observable λ_t . Extending this to the hidden-regime setting — bounding the additional error from regime inference — is an open theoretical question.

7 Conclusion

This project extends the state-of-the-art ST-Hedging framework to markets with hidden liquidity regimes. The proposed **Regime-Aware Residual Policy** integrates a GRU belief encoder, an asymptotic base rate computed from belief-weighted parameters, a neural residual corrector, and a stress-sensitive liquidity gate into a unified end-to-end trainable architecture.

Experimentally, the model achieves a mean utility of **+0.001041** — more than double the Deep Hedging baseline and significantly above all formula-based benchmarks. It reduces terminal tracking MSE by 72% relative to Deep Hedging and approaches Oracle performance (which requires knowledge of the true hidden state) in tracking accuracy.

The key insight is that regime-adaptive behaviour — both in parameter estimation (through belief-weighted effective parameters) and in risk management (through the stress liquidity gate) — is essential for robust hedging performance in realistic markets. The base paper’s ST-Hedging framework provides an excellent foundation, and the residual-on-asymptotic design ensures fast convergence and stable training even under the additional complexity of hidden regimes.

Reproducibility. All code is provided. Experiments were run on CPU with PyTorch 2.x. Training time is approximately 2–3 minutes for 300 epochs on a modern laptop CPU. The default random seed is 42.

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